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Why Carbon Pricing Isn't Working

Good Idea in Theory, Failing in Practice

Jeffrey Ball

For decades, as the reality of climate change has set in, policy-makers have pushed for an elegant solution: carbon pricing, a system that forces polluters to pay when they emit carbon dioxide and other greenhouse gases. Among the places that have imposed or scheduled it are Canada, China, South Korea, the EU, and about a dozen U.S. states. Much as a town charges people for every pound of trash tossed into its dump, these jurisdictions are charging polluters for every ton of carbon coughed into the global atmosphere, thus encouraging the dirty to go clean.

In theory, a price on carbon makes sense. It incentivizes a shift to low-carbon technologies and lets the market decide which ones will generate the biggest environmental bang for the buck. Because the system harnesses the market to help the planet, it has garnered endorsements across the political spectrum. Its adherents include Greenpeace and ExxonMobil, leftist Democrats and conservative Republicans, rich nations and poor nations, Silicon Valley and the Rust Belt. Essentially every major multilateral institution endorses carbon pricing: the International Monetary Fund, the UN, and the World Bank, to name a few. Christine Lagarde, the managing director of the IMF, spoke for many in 2017 when she recommended a simple approach to dealing with carbon dioxide: "Price it right, tax it smart, do it now."

In practice, however, there's a problem with the idea of slashing carbon emissions by putting a price on them: it isn't doing much about climate change. More governments than ever are imposing prices on

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carbon, even as U.S. President Donald Trump backpedals on efforts to combat global warming, yet more carbon than ever is wafting up into the air. Last year, the world's energy-related greenhouse gas output, which had been flat for three years, rose to an all-time high. Absent effective new policies, the International Energy Agency has projected, energy-related greenhouse gas emissions will continue rising through at least 2040.

If governments proved willing to impose carbon prices that were sufficiently high and affected a broad enough swath of the economy, those prices could make a real environmental difference. But political concerns have kept governments from doing so, resulting in carbon prices that are too low and too narrowly applied to meaningfully curb emissions. The existing carbon-pricing schemes tend to squeeze only certain sectors of the economy, leaving others essentially free to pollute. And even in those sectors in which carbon pricing might have a significant effect, policymakers have lacked the spine to impose a high enough price. The result is that a policy prescription widely billed as a panacea is acting as a narcotic. It's giving politicians and the public the warm feeling that they're fighting climate change even as the problem continues to grow.

Sometime this century, global temperatures are all but certain to cross what scientists warn is a perilous threshold: two degrees Celsius above their preindustrial levels. The two-degree line, a notion introduced in 1975 by the economist William Nordhaus, is less an environmental cliff than a political rallying cry. But beyond it, a range of problems will grow worse, including extreme weather events, coastal flooding, and, in tropical and temperate regions, a reduction in the yields of crucial crops such as wheat and rice. So the world needs solutions that do more than merely chip away at the problem. What's required are more targeted moves—ones that are politically difficult but possible and environmentally effective. These include phasing out coal as a fuel for electricity, except where coal is paired with technology to capture its carbon emissions; keeping nuclear power plants up and running; slashing fossil fuel subsidies; raising gasoline taxes; reducing the cost of renewable power; and toughening energy-efficiency requirements.

Carbon pricing need not be abandoned. It can, at least at the margins and in concert with these more direct carbon-cutting policies, help channel money into cleaner energy options. But there is little evidence

for what has become an article of faith in the climate fight: that carbon pricing should be society's main tool to keep the planet cool.

PERMISSION TO POLLUTE

The roots of the notion of curbing pollution by pricing it go back nearly a century. In 1920, the British economist Arthur Pigou developed the concept of an economic “externality”: a benefit or cost that is not priced into a given activity but can be, through what would come to be called a Pigouvian tax. Nearly 50 years later, in the late 1960s, two economists working separately—Thomas Crocker and John Dales—proposed a different sort of pricing mechanism to limit emissions: a combination of government-mandated caps and tradable emission allowances, a one-two punch that would come to be known as “cap and trade.”

Under a cap-and-trade system, a government imposes a limit on the amount of carbon that the economy, or specified sectors of it, may emit. It apportions responsibility for curbing emissions in line with that cap to individual players, such as companies. At the same time, it creates a tradable currency called a carbon permit; each permit allows its bearer to emit one metric ton of carbon dioxide. In some cap-and-trade systems, the original permits are given away for free, whereas in others, they are sold—creating revenue for governments. If a polluter's expected emissions exceed the cap, it must either curb its emissions—say, by installing more efficient manufacturing equipment or shifting to cleaner energy sources—or buy more permits on the market. A polluter whose emissions are trending below its cap can sell its excess permits on the market. In some systems, the market alone sets the price; in others, the government imposes a floor and a ceiling on the permit price.

The basic idea behind a cap-and-trade system is twofold. First, by forcing polluters to pay for the carbon they emit, the system incentivizes them to invest in lower-carbon solutions, thus directing more private capital—and, in turn, more research and innovation—toward clean technology. Second, by spreading the burden for cutting carbon across an entire sector—or, ideally, across an entire economy—the system helps each regulated player find the lowest-cost way to reduce its carbon output.

The first major use of emission trading was in the United States, to fight local air pollution. The federal government used it to phase



Bad atmospherics: smokestacks in Jilin, China, February 2013

out leaded gasoline starting in the 1980s and to combat acid rain, an effect of power plant emissions, starting in the early 1990s. Both campaigns succeeded, but limiting pollution from tailpipes and smokestacks in a single city or region is infinitely easier than slashing emissions of invisible carbon dioxide around the planet.

Carbon pricing started in the 1990s in Scandinavia and expanded in the following decade throughout Europe. More recently, it has taken hold in California, the Northeast of the United States, much of Canada, and many other places. Today, according to the World Bank, 42 countries and 25 subnational jurisdictions—together representing about half of global GDP and a quarter of global greenhouse gas emissions—have imposed or are pursuing a price on carbon, through either a cap-and-trade system or a carbon tax.

But because many jurisdictions have imposed carbon prices just in certain sectors of their economies, carbon pricing covers only about 15 percent of global emissions, the World Bank has calculated. That portion should grow to between 20 and 25 percent once China, the world's largest carbon emitter, implements a nationwide carbon-pricing program, as it has promised to do. Yet even that share would fall far short of the 50 percent of global emissions that a World Bank panel has said needs to be covered by carbon pricing within a decade in order to meet the global carbon-reduction goals set forth in the Paris climate accord.

Why does carbon pricing squeeze certain sectors more than others? The answer is that it works well for industries that use a lot of fossil energy, that have technologies available to them to reduce that energy use, and that can't easily relocate to places where energy is cheaper. In other words, it works well in the power and heating sector, which produces about 25 percent of global emissions. That industry is

Carbon pricing covers only about 15 percent of global emissions.

dominated by localized utilities that can curb their carbon emissions in a number of ways: by switching to more efficient equipment for burning fossil fuels, by shifting from higher-carbon fossil fuels such as coal to lower-carbon

ones such as natural gas, by increasing their use of renewable energy, by capturing the carbon dioxide they produce and sequestering it, or by incentivizing their customers to waste less electricity.

Carbon pricing tends not to work well for curbing emissions from buildings, which generate about six percent of global emissions. Builders rarely occupy the buildings they build, which means they don't pay the energy bills and thus have little incentive to foot the capital cost of more efficient buildings. Nor does carbon pricing work well to curb emissions from transportation, which account for about 14 percent of the global total. Studies show that drivers are usually unresponsive to modest increases in gasoline and diesel taxes. And although they do respond to big hikes, taxes that high tend to be political nonstarters. No wonder, then, that carbon-pricing regimes tend not to tamp down emissions from buildings and vehicles.

Just as the breadth of a carbon-pricing system matters, so does the price it puts on each metric ton of carbon dioxide. In 2017, a group of leading economists known as the High-Level Commission on Carbon Prices concluded that carbon prices would have to be between \$40 and \$80 per metric ton by 2020, and between \$50 and \$100 by 2030, to achieve the emission cuts called for in the Paris climate accord. (Even in the unlikely event that the 195 nations that have agreed under the accord to voluntarily constrain their carbon outputs met their promises, that wouldn't stop global temperatures from surpassing the two-degree threshold.) But of the global emissions now subject to a carbon price, just one percent are priced at or above the commission's \$40 floor of ecological relevance. Three-quarters are priced below \$10. The upshot: more

than two years after the ostensible watershed moment of Paris, a mere 0.15 percent of global greenhouse gas emissions are subject to a carbon price that economists deem high enough to make much of an environmental difference.

Four countries have priced carbon at or above that \$40 floor, according to the World Bank: Finland, Liechtenstein, Sweden, and Switzerland. These are rich nations with a deep-seated culture of environmental protection. They also have, by global standards, comparatively low-carbon electricity systems, thanks in large part to plentiful hydropower and, in the cases of Finland, Sweden, and Switzerland, a great deal of nuclear power, too. All told, they couldn't be more different from the sorts of places—China, India, Africa, and the rest of the developing world—that most matter in the fight against climate change.

The same is true of most of the U.S. states, including California, Maine, New York, and Vermont, that have chosen to price at least some of their carbon either on their own or through a regional cap-and-trade program for power plant emissions. Compared with other U.S. states, these tend to have ample solar power, wind power, or hydropower, and they are less reliant on high-carbon coal.

It's not just governments that are joining the carbon-pricing stampede. More than 1,400 companies globally, including some of the world's largest multinationals, are voluntarily integrating carbon prices into their investment decisions, according to CDP, a nonprofit that gathers environmental data from companies and governments. When, say, an oil company decides whether to drill in a certain field or a bank decides whether to loan to a certain project, it first tries to calculate what would happen to its profits if the government imposed a particular carbon price. In theory, doing this should lead companies to favor less carbon-intensive investments.

Here, too, however, the reality is underwhelming. To decarbonize the energy system enough to meet even the limited goals set in Paris, annual global investment in low-carbon technologies would have to rise by about \$700 billion by 2030, according to the World Bank. The bank also estimates that an international carbon market could incentivize about one-third of that—about \$220 billion annually. That figure in itself is telling: even under the rosiest of circumstances, carbon pricing will produce only a fraction of the emission cuts needed to put the world onto a sufficiently low-carbon path.

THE PRICE IS WRONG

How a strategy so widely seen as so promising has failed to live up to its ideal is a tale of good intentions thwarted by economic and political realities. Europe's experience is instructive. Launched in 2005, the EU's emission-trading system was designed to cover electricity generators and energy-intensive industries such as cement and steel manufacturing. But from the beginning, the companies the system covered got plentiful free permits. That was a compromise EU officials made to mollify opposition from industry. It meant that only those companies that experienced unexpected rises in emissions had to pay much for the right to pollute.

When the 2008 global financial crisis struck, European economic activity declined, and so did emissions. Companies found themselves with more free permits than they needed, and European carbon prices tanked, from more than 25 euros per metric ton in 2008 to less than five euros in 2013. In recent years, the EU has toughened the system somewhat; among other things, it has required more companies to buy more of their permits, and it has broadened the system to cover airline flights within the EU. But the permits remain so cheap that the program is not prodding emission reductions in line with the long-term carbon-reduction goals that it has set. Between 2015 and 2016, EU emissions fell by 0.7 percent across the bloc—enough to keep the EU on track to meet its goal of cutting emissions to 20 percent below 1990 levels by 2020, but not enough, officials have admitted, to meet the EU's more ambitious commitment of reducing them to 80 percent below 1990 levels by 2050. And in 2017, emissions covered by the EU's carbon-pricing system actually rose, for the first time in seven years, the result of stronger than expected industrial output.

Last year, recognizing significant flaws in its carbon-pricing system, the EU agreed to redesign it. The new version, set to take effect in 2021, seeks to tighten emission limits, reduce handouts of free permits, and pull excess permits off the market if their price falls below a certain level. But the reforms are probably too little, too late. The price of permits has risen markedly this year, from about eight euros in January to about 14 euros in mid-May. Nevertheless, some analysts have predicted that their price will average only about 18 euros per metric ton in 2020, about half the price that the World Bank says will be necessary to make a real dent in carbon emissions. In a November 2017 report, the Mercator Research Institute on Global

Commons and Climate Change, a Berlin-based organization, cited persistently low permit prices when it warned that the EU's carbon-pricing system is "in a crisis."

California, the world's sixth-largest economy, has had similar problems. Although it produces only about one percent of global greenhouse gas emissions, it has long been a bellwether for environmental policy, imposing regulations that are later adopted across the country and around the world. The state launched its cap-and-trade system for carbon in 2012, part of a broader plan to cut its emissions to 1990 levels by 2020—a goal less ambitious than the EU's but more ambitious than the U.S. federal government's. California is all but sure to meet that target. But even though emissions from power generation covered by its cap-and-trade system fell in 2016, those related to transportation—the state's biggest source of carbon emissions—rose that year. What's more, as an analysis released last year by Near Zero, a nonprofit research group in California, concluded, the decline in power plant emissions owes little to carbon pricing. Instead, it is largely the product of an increased use of hydropower (a result of higher rainfall) and a greater production of wind and solar power (a result of state renewable energy mandates). As of mid-May, California's carbon price was around \$15 per metric ton. It was that low because factors other than the carbon market led power producers to curb their emissions, leaving companies with extra permits that they had gotten from the state for free.

Like Europe, California is moving to add more bite to its carbon-pricing system. It wants to force far deeper emission cuts, in line with the EU's ambitions: to 40 percent below 1990 levels by 2030 and 80 percent below 1990 levels by 2050. A plan now under consideration could increase the carbon price to between \$81 and \$150 per metric ton in 2030. If such a higher price materializes, it should spur big cuts in emissions. But the state has yet to decide on the proposed plan, and the fight is intense. In public hearings and through private lobbying, oil producers and power companies are sparring with environmental groups. In March, an official from Pacific Gas and Electric, California's largest utility, told state officials that a 2030 carbon price of \$150, a level that some environmentalists call sensible, would be "very high" and would not "strike that appropriate balance" between planet and pocketbook.

California's revised system would reflect a new carbon-pricing approach that is drawing bipartisan support and interest from policymakers. Called a "revenue-neutral" carbon price or a "carbon dividend,"

this scheme returns to consumers some or all of the money raised by the selling of permits rather than putting that revenue into government coffers. The allure of this approach is that although it still forces big emitters to pay, goading them to pollute less, it returns revenue to consumers (for instance, as tax rebates), compensating them for the higher prices they have to pay for energy and other goods as a result of the price on carbon.

In theory, returning to consumers money raised from a carbon price should be popular, giving policymakers political cover to impose a carbon price high enough to make a difference on climate change. But in reality, even this idea faces opposition from interests that would be hit hardest by the carbon price. British Columbia implemented a revenue-neutral carbon price in 2008 and initially saw its emissions drop. But in 2012, amid political blowback, the province froze its carbon price, at 30 Canadian dollars per metric ton. Unsurprisingly, emissions started rising again. This spring, British Columbia raised the carbon price to 35 Canadian dollars per metric ton—lower than a government advisory panel suggested was necessary.

THE CHINESE DREAM

China, the world's factory floor and most populous country, is the most important piece in the climate change puzzle. Unless it slashes its carbon emissions, little that the rest of the world does in the climate fight will matter much. It is the world's largest producer of both coal-fired power and renewable energy. And with its powerful central government, it would seem uniquely able to execute a carbon-pricing revolution. In 2013 and 2014, after studying the European and Californian examples, China launched carbon-pricing tests in five cities and two provinces. And in 2015, Chinese President Xi Jinping announced with great fanfare that China would soon take carbon pricing nationwide. Trading is expected to start in 2019 or 2020.

As has been the case elsewhere, however, carbon pricing is unlikely to reduce carbon emissions dramatically in China. Those emissions are expected to peak between 2025 and 2030. That might seem like good news, but it's not good enough. The start of a decline in carbon output from the world's biggest emitter won't fix climate change; what's necessary is for total global emissions to plummet. Moreover, assuming that China's emissions do in fact peak, which seems likely, they will do so in response to broad changes in the economy that

have next to nothing to do with a price on carbon. Those changes—in particular, improvements in the energy efficiency of manufacturing and reductions in emissions from coal-fired power plants—will be driven primarily by economic and public health priorities. Indeed, in the pilot programs that China has rolled out in various localities, carbon in mid-May was trading at between about \$2 and \$9 per metric ton, too low to meaningfully change the behavior of companies or citizens.

None of this is terribly surprising, given that the plans for China's nationwide carbon-pricing system have been steadily watered down. The scheme was originally designed to cover between 6,000 and 7,000 companies across multiple industries. Instead, it will, at least initially, cover 1,700 power producers. China may soften its carbon-pricing system even further. The government has yet to decide how many permits it will provide to companies, and it could choose to hand them out for free. Already, companies that successfully lobbied to receive free additional emission permits under the pilot programs are pushing for the right to use those permits under the nationwide system. This has the potential to create an oversupply of permits in China similar to the ones that have contributed to the low prices in the EU and California. As one carbon-pricing expert involved in the design of China's system told me, "We are repeating the same mistakes that the EU market and California have done."

BLUNT TOOLS

For all its shortcomings, carbon pricing has done two important things. It has accustomed powerful economic players—governments, companies, and, to a lesser extent, consumers—to the notion that they will have to integrate decarbonization into their spending decisions. In the process, it has prodded those actors to put more effort into discovering both the technologies and the business models that would most cost-effectively cut carbon emissions to an environmentally meaningful extent. But carbon pricing is failing to produce emission cuts that are significant—and the time for tinkering is running out. Because carbon pricing is giving humanity the illusion that it is dealing responsibly with climate change, it is reducing the pressure to adopt other carbon-cutting measures, ones that would hit certain sectors harder and that would produce faster reductions.

Seriously addressing climate change in the immediate future demands not a theoretically effective strategy but an actually effective one.

That's because with each passing year, more carbon accumulates in the atmosphere, and more global warming becomes inevitable. Slashing emissions in the near term is crucial. But in 2017, global energy-related carbon emissions rose for the first time in four years. The 1.4 percent rise was due to an increase in coal use, particularly in Asia, and to a slowdown in worldwide energy-efficiency improvements, the result of cheap fossil fuel.

Since carbon pricing on its own is not reversing that trend, what else is needed? Policymakers should start with electricity, arguably the easiest sector to clean up, owing to the ready availability of natural gas and increasingly cost-effective renewable energy sources. Where

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feasible, coal, the most carbon-intensive fossil fuel, should be phased out by fiat unless technology to mitigate its emissions—technology known as “carbon capture and storage,” or CCS—can be scaled up. But make no mistake: coal is all but certain to remain a major electricity source for decades, particularly in the developing world. China

and India sit on massive supplies of it, and even as both countries rapidly scale up renewable power from a tiny base, they will be hard-pressed to get rid of coal anytime soon. In the meantime, then, the imperative is to resolve the technological, legal, and political impediments to CCS.

Finding an economically and politically viable way to capture and store carbon from fossil fuel consumption is crucial not just for electricity production but also for industrial processes such as cement and steel production. These activities emit huge quantities of carbon dioxide, and for now, there is no viable way to power them other than by burning fossil fuels. But efforts to develop CCS technology have stalled as carbon pricing has floundered, because absent a strong government push to reduce carbon emissions, companies have no reason to spend money on it. Experts estimate that a carbon price well above \$100 per metric ton, and perhaps much higher, would be needed to create enough of an incentive for firms to invest in large-scale CCS. Given that a carbon price that high anytime soon seems to be a pipe dream, governments will have to provide more direct financial support for the technology.

Meanwhile, humanity cannot afford to reject nuclear power, a reliable, carbon-free energy source. The safety and proliferation concerns about nuclear power are real, but they can be mitigated through a combination of newer nuclear technologies and smarter regulations. Given public opposition to nuclear power, and given the declining cost of renewable energy, nuclear power's share of global electricity generation is expected to remain relatively flat. Even so, shutting down nuclear plants that have years of life left in them, as Germany, Japan, California, and other U.S. states are doing, represents a step backward for the climate.

Policymakers will also have to figure out how to unlock the potential of renewable energy. The cost of wind and solar power is plummeting, but it is still too high, and these sources remain a small slice of the total energy supply. To slash costs further, policymakers should, for example, resist the temptation to impose protectionist policies, such as tariffs on imported renewable energy equipment, which only make renewable energy more expensive.

Compared with the electricity sector, transportation is harder to decarbonize. True, electric cars will likely proliferate as their cost continues to fall, and if powered by clean electricity, they could become a major climate-fighting tool. But batteries remain too expensive, and it will likely take decades to replace the fleet of vehicles already on the road. So oil will, according to most projections, continue to power most transportation until the middle of the century and perhaps well beyond it. For the foreseeable future, then, the key is to minimize the wasteful consumption of oil.

One important way to do that is to raise the price of gasoline and diesel fuel. In developed countries, particularly in the United States, that means raising the price at the pump through taxes. In developing countries, that means rolling back motor fuel subsidies. That is politically difficult. But governments from Mexico to Saudi Arabia are showing it's possible.

Then there are improvements in energy efficiency that can be made to buildings, appliances, vehicles, and aircraft. The payoff of such improvements remains an open question; there is evidence that as a given thing's energy efficiency improves, people tend to use that thing more, negating any reduction in carbon emitted. That said, efficiency improvements are an important factor in decreasing carbon emissions. Rules forcing greater energy efficiency—particularly in buildings and cars—work.

MOVING ON

Humanity has solved a host of important environmental problems—once it decided those problems were crises. Crushing smog in postwar Los Angeles helped spur the 1970 Clean Air Act. When the Cuyahoga River in Cleveland, then strewn with industrial waste, burst into flames in 1969, another in a line of river fires, that hastened the Clean Water Act of 1972. Public worry in the 1980s about the growing ozone hole led to the 1987 Montreal Protocol, which started the phaseout of ozone-depleting chemicals. But climate change, this century's grand challenge, is different from these past problems. It is not just more serious; it is also massively harder to solve. Physics, politics, and economics all conspire to make climate change what social scientists call a "wicked problem"—one in which every supposed solution creates another complication.

That does not, however, necessarily mean that climate change cannot be tamed. Although the planet is all but certain to cross the two-degree threshold, minimizing greater warming is both possible and pressing. Phasing out high-carbon coal, speeding the development of CCS, maintaining nuclear energy, slashing renewable energy costs, and raising fuel prices would make a difference. So would ratcheting up efforts unrelated to energy, such as combating deforestation. To be sure, such a grab bag of policies lacks the intellectual tidiness of a carbon price. Some of the policies will be hard to achieve; others will fail. And all would be helped by an effective carbon price. But pursuing these measures directly offers a politically realistic path to significant environmental benefit.

Maybe one day carbon pricing will be the best tool for fighting climate change. But the planet doesn't have time to wait. To the extent that the carbon-pricing experiment lets policymakers and the public delude themselves that they are meaningfully addressing global warming, it's not just ineffectual; it's counterproductive. The time has come to acknowledge that this elegant solution isn't solving the problem it was designed to solve. In the toughest environmental fight the world has ever faced, a good idea that isn't working isn't good enough. 🌍